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COMPREHENSIVE STUDY GUIDE

Pharmaceutical Chemistry

Foundations + PCH 202 Assays & Determination

PART ONE · Foundational Pharmaceutical Chemistry

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Periodic trends · bonding · hybridization · reaction mechanisms
Functional groups · IUPAC nomenclature · coordination & chelation

PART TWO · PCH 202 — Pharmaceutical Assays & Determination

QA · QC · GMP · assay types · errors, precision & accuracy
Physicochemical techniques · chromatography · immunoassay
Microbiological assay · bioassay · concentration units & purity

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Everything a pharmacist needs to know about how a drug behaves - whether it ionises, dissolves in water or fat, reacts, or binds a receptor - ultimately traces back to how electrons are arranged around atoms. That arrangement, called the **electronic structure**, is governed not by classical physics but by **quantum mechanics**, the framework that describes matter at the scale of individual particles. This opening section builds the conceptual foundation on which every later topic rests.

1.1 Why classical physics is not enough

Electrons are far too small and fast for Newtonian mechanics to describe accurately. Instead, their behaviour is captured by a **wavefunction** - a mathematical expression whose square gives the probability of finding an electron at a particular point in space. Solving for these wavefunctions is what generates the familiar shells, subshells and orbitals of chemistry.

1.2 The models and equations you should know

- » **Non-relativistic systems** are handled by the **Schrodinger equation**, written compactly as $H\psi = E\psi$, where H is the Hamiltonian (total-energy) operator, ψ is the wavefunction and E is the allowed energy (an eigenvalue). It works well for lighter atoms.
- » **Relativistic systems** - very heavy atoms whose inner electrons approach the speed of light - require **Dirac's equation** instead, which folds relativity into quantum mechanics.
- » The **Born-Oppenheimer approximation** makes molecules tractable by assuming the heavy, sluggish nuclei are effectively frozen while the light, fast electrons rearrange around them, letting nuclear and electronic motion be solved separately.
- » The **Hartree-Fock method** is an iterative ('self-consistent field') technique for approximating the wavefunctions and energies of many-electron systems, accounting for electron-electron repulsion in an averaged way.

KEY POINT

The single most important outcome of solving the Schrodinger equation for an atom is that energy, shape and orientation of orbitals cannot take just any value - they are **quantised**. This restriction is exactly what gives rise to the four quantum numbers examined in Section 2.

Each electron in an atom is uniquely described by a set of four **quantum numbers**. Together they specify the size, shape, spatial orientation and spin of the orbital the electron occupies. The **Pauli exclusion principle** guarantees that no two electrons in the same atom share an identical set of all four - which is why orbitals fill in the orderly way they do.

2.1 Principal quantum number (n)

This defines the main energy level, or **shell**, and therefore how far, on average, the electron sits from the nucleus. It takes positive-integer values ($n = 1, 2, 3, \dots$); larger n means higher energy and a bigger orbital. The maximum number of electrons a shell can hold is given by $2n^2$ - so 2 electrons for $n=1$, 8 for $n=2$, 18 for $n=3$, and so on.

2.2 Azimuthal (angular-momentum) quantum number (l)

This fixes the **shape** of the orbital and identifies the subshell. For a given shell it runs from 0 up to (n-1). Each value corresponds to a lettered subshell with a characteristic shape.

l value	Subshell	Orbital shape	Orientations
0	s	Spherical	1
1	p	Dumbbell	3 (p_x, p_y, p_z)
2	d	Cloverleaf	5
3	f	Complex	7

2.3 Magnetic quantum number (m_l)

This describes how an orbital is **oriented** in three-dimensional space relative to an external magnetic field. For a given l it ranges from -l through 0 to +l, giving $(2l + 1)$ values - which is precisely the number of orbitals in that subshell. For example $l = 1$ gives $m_l = -1, 0, +1$, the three p-orbitals.

2.4 Spin quantum number (m_s)

This captures the electron's intrinsic **spin**, which can be $+\frac{1}{2}$ or $-\frac{1}{2}$ (loosely, 'clockwise' or 'anticlockwise'). Because an orbital can hold at most two electrons, those two must have opposite spins - the microscopic basis of electron pairing and of magnetic behaviour.

Quantum number	Symbol	Determines	Possible values
Principal	n	Energy level / shell	1, 2, 3, ... (positive integers)
Azimuthal	l	Shape of the orbital	0 to (n - 1)
Magnetic	m_l	Orientation in space	-l to +l (including 0)
Spin	m_s	Direction of electron spin	$+\frac{1}{2}$ or $-\frac{1}{2}$

NOTE

No two electrons in one atom can carry the same four quantum numbers (Pauli). This is not an arbitrary rule - it is what forces electrons into successive orbitals and ultimately builds up the entire periodic table.

§3 ELECTRONIC CONFIGURATION OF ELEMENTS

An element's **electronic configuration** is the specific way its electrons are distributed across shells, subshells and orbitals. Because valence (outer) electrons dictate reactivity, configuration is the key that unlocks periodic trends and bonding. Configurations are written using the shell number, the subshell letter and a superscript count - for example oxygen is $1s^2 2s^2 2p^4$.

3.1 The three governing rules

- » **Aufbau principle** - electrons fill the lowest-energy orbital available before moving to higher ones (build-up from the bottom).
- » **Pauli exclusion principle** - an orbital holds a maximum of two electrons, and they must have opposite spins.
- » **Hund's rule** - within a set of equal-energy (degenerate) orbitals, electrons occupy them singly with parallel spins before any pairing begins, which minimises repulsion and maximises stability.

3.2 Order of orbital filling (Madelung / diagonal rule)

Orbitals do not simply fill in numerical order; sub-levels overlap in energy. The experimentally observed sequence is:



3.3 Shell capacities and worked examples

Shell (n)	Subshells available	Maximum electrons
1	1s	2
2	2s, 2p	8
3	3s, 3p, 3d	18
4	4s, 4p, 4d, 4f	32

Subshell	Orbitals	Max electrons
s	1	2
p	3	6
d	5	10
f	7	14

Worked examples. Carbon ($Z=6$): $1s^2 2s^2 2p^2$. By Hund's rule its two 2p electrons occupy separate orbitals with parallel spins rather than pairing. Oxygen ($Z=8$): $1s^2 2s^2 2p^4$ - the 2p set fills as up, up, up-down. Calcium ($Z=20$): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$, showing the 4s filling before 3d.

PHARMACEUTICAL RELEVANCE

Valence electrons - those in the outermost shell - are what determine an element's chemistry. Sodium ($1s^2 2s^2 2p^6 3s^1$) has a single, easily lost valence electron, which is exactly why it is so reactive and why sodium salts are so common in pharmacy.

§4 THE PERIODIC TABLE & PERIODIC TRENDS

The modern periodic law states that the properties of the elements are a periodic function of their **atomic numbers** (not atomic masses, as Mendeleev and Meyer originally supposed). Arranging elements by increasing atomic number places those with similar valence configurations - and therefore similar chemistry - into the same vertical group.

4.1 How the table is organised

- » **Blocks** - the table divides into the s-block, p-block, d-block (transition metals) and f-block (lanthanides and actinides), named after the subshell that the valence electrons are entering.
- » **Types of element** - metals occupy the left and centre, non-metals the upper right, and metalloids form the staircase boundary between them.
- » **Element data** - each tile carries the symbol, atomic number, atomic mass and often the name.

4.2 The key periodic trends

Trend	Across a period (L to R)	Down a group	Underlying reason
Atomic radius / size	Decreases	Increases	More shells down a group; higher nuclear pull across a period
Ionisation energy	Increases	Decreases	Outer electrons held more tightly as nuclear charge rises
Electron affinity	Generally increases	Decreases	Smaller atoms attract an added electron more strongly (F is an exception)
Electronegativity	Increases	Decreases (except Group VIIIA)	Reflects an atom's pull on a shared bonding pair

Electronegativity deserves special attention: the larger the difference in electronegativity between two bonded atoms, the more polar - or ultimately ionic - the bond becomes. This single idea links atomic structure directly to the bonding behaviour covered next.

§5 CHEMICAL BONDING

Atoms combine because doing so **lowers their energy** and confers stability, typically by achieving a full outer shell of eight electrons (the **octet rule**, configuration $ns^2 np^6$). Which of the three bond types forms depends on how the valence electrons are shared or transferred, which in turn is governed by electronegativity.

5.1 What drives bond formation

- » **Octet rule** - atoms gain, lose or share electrons to reach eight in the outer shell.
- » **Electronegativity** - the greater the difference between two atoms, the more the electrons shift toward one of them.
- » **Energy consideration** - a bond forms only if the resulting molecule is lower in energy, and therefore more stable, than the separate atoms.

5.2 Comparing the three bond types

Bond type	Electron behaviour	Forms between	Examples
Covalent	One or more pairs of electrons shared	Non-metals of similar electronegativity	H_2O , O_2 , CH_4
Ionic	Electrons transferred, creating attracting ions	A metal and a non-metal	$NaCl$, CaO , MgF_2
Coordinate (dative)	Shared pair donated entirely by one atom	An electron-pair donor and an electron-deficient acceptor	NH_4^+ , CO , Al_2Cl_6

5.3 Covalent bonds in more detail

A covalent bond is the sharing of one or more electron pairs, typically between non-metals. Bonds may be **single** (one shared pair, e.g. H_2 , CH_4), **double** (two pairs, e.g. O_2 , CO_2) or **triple** (three pairs, e.g. N_2 , C_2H_2). Covalent compounds tend to have low melting and boiling points, conduct electricity poorly (graphite excepted), and dissolve in non-polar solvents.

5.4 Ionic bonds in more detail

An ionic bond is the electrostatic attraction between oppositely charged ions produced when a metal donates electrons (forming a cation, e.g. Na^+ , Ca^{2+}) and a non-metal accepts them (forming an anion, e.g. Cl^- , O^{2-}). The strong electrostatic lattice gives ionic compounds high melting and boiling points, water solubility, and

electrical conductivity when molten or dissolved.

5.5 Coordinate (dative) bonds in more detail

A coordinate bond is a covalent bond in which **both** shared electrons come from a single atom - a donor with a lone pair - given to an electron-deficient acceptor. Once formed it is indistinguishable in strength from an ordinary covalent bond. Classic examples are the ammonium ion (NH_3 donating a lone pair to H^+) and carbon monoxide. Coordinate bonding is the very basis of the metal-ligand complexes discussed in Section 12.

§6 ORBITALS & HYBRIDIZATION

An **orbital** is a region around a nucleus where an electron is most likely to be found. When atoms bond, their atomic orbitals combine into **molecular orbitals**, and in many molecules the atomic orbitals of the central atom first reorganise into **hybrid orbitals** that explain the observed shapes. This section connects orbital theory to real molecular geometry.

6.1 Atomic vs molecular orbitals

Feature	Atomic orbitals	Molecular orbitals
Belong to	A single atom	The whole molecule
Formation	Defined by n, l, m_l	Overlap of atomic orbitals (LCAO)
Number	Set by the quantum numbers	Equals the number of combining atomic orbitals
Examples	1s, 2p of hydrogen	σ and π orbitals in H_2 , O_2

6.2 Molecular-orbital theory (MOT)

When atomic orbitals overlap they produce two kinds of molecular orbital: a lower-energy, stabilising **bonding orbital** (constructive overlap, e.g. σ , π) and a higher-energy, destabilising **antibonding orbital** (destructive overlap, denoted σ^* and π^*). The core principles are that the number of molecular orbitals equals the number of atomic orbitals combined, bonding orbitals lie below antibonding ones, and electrons fill them following the Aufbau principle and Hund's rule. Notably, the molecular-orbital picture of O_2 predicts unpaired electrons in its π^* orbitals, correctly explaining why oxygen is **paramagnetic** - something simple Lewis structures cannot.

6.3 Hybridization and molecular shape

Hybridization is the mixing of atomic orbitals on one atom to form new, equal-energy orbitals that point in the directions needed for bonding. It explains the geometries predicted by **VSEPR** theory and the bond angles actually measured.

Hybridization	Orbitals mixed	Hybrid orbitals	Geometry	Bond angle	Examples
sp	1s + 1p	2	Linear	180°	BeCl_2 , CO_2 , C_2H_2
sp ²	1s + 2p	3	Trigonal planar	120°	BF_3 , C_2H_4
sp ³	1s + 3p	4	Tetrahedral	109.5°	CH_4 , NH_3 , H_2O
sp ³ d	1s + 3p + 1d	5	Trigonal bipyramidal	90° & 120°	PCl_5
sp ³ d ²	1s + 3p + 2d	6	Octahedral	90°	SF_6

NOTE

Watch the effect of lone pairs: nitrogen in NH_3 and oxygen in H_2O are both sp^3 -hybridized like carbon in CH_4 , but their lone pairs push the bonding pairs closer, giving trigonal-pyramidal and bent shapes respectively rather than a perfect tetrahedron.

§7 BONDING TYPES & ELECTRONIC EFFECTS

Beyond simply classifying bonds, a pharmacist must understand the finer electronic features that control a molecule's strength, reactivity, polarity and stability. Four ideas do most of the work: sigma bonds, pi bonds, bond polarity and resonance.

7.1 Sigma (σ) and pi (π) bonds

- » **Sigma bond** - the strongest covalent bond, formed by direct head-on (axial) overlap of orbitals (s-s, s-p or p-p). Every single bond is a sigma bond, and it permits free rotation about the bond axis.
- » **Pi bond** - a weaker bond formed by sideways (lateral) overlap of p-orbitals. Pi bonds appear only in double and triple bonds (always alongside one sigma bond), restrict rotation, and are the reactive feature of alkenes, alkynes and aromatic systems.

A carbon-carbon double bond (as in ethene) is therefore one σ plus one π bond; a triple bond (as in ethyne) is one σ plus two π bonds.

7.2 Bond polarity

When two bonded atoms differ in electronegativity, the shared electrons are pulled toward the more electronegative atom, creating a **dipole** with partial charges ($\delta+$ and $\delta-$). Bonds are classified as **non-polar covalent** (equal sharing, e.g. O_2 , N_2), **polar covalent** (unequal sharing, e.g. H_2O , NH_3 , HF) or fully **ionic** (complete transfer, e.g. NaCl). Polarity governs a drug's water solubility, its ability to hydrogen-bond, and how it interacts with receptors.

7.3 Resonance

Resonance is the **delocalisation** of electrons across a molecule when more than one valid Lewis structure can be drawn. The true molecule is a hybrid of these structures - none is 'real' on its own. Delocalisation lowers energy and increases stability; the classic example is benzene, whose six carbon-carbon bonds are all identical in length because its pi electrons are spread evenly around the ring.

§8 BOND FORMATION, FISSION & REACTION MECHANISMS

Chemical reactions are, at heart, the making and breaking of bonds. Understanding **how** bonds break - and what intermediates result - lets us predict reaction pathways, a skill central to understanding drug synthesis and drug degradation alike.

8.1 Bond fission: two ways a bond can break

	Homolytic fission	Heterolytic fission
What happens	Bond splits evenly; each atom keeps one electron	Bond splits unevenly; one atom takes both electrons
Products	Neutral free radicals ($\text{A}\cdot + \text{B}\cdot$)	Ions - a cation and an anion ($\text{A}^+ + \text{B}^-$)

	Homolytic fission	Heterolytic fission
Favoured by	Non-polar bonds; UV light or heat	Polar bonds; polar solvents
Typical setting	Radical reactions, e.g. $\text{Cl}_2 \rightarrow 2 \text{Cl}\cdot$ under UV	Acid-base and ionic reactions, e.g. $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ in water

8.2 Common reaction mechanisms

- » **Radical reactions** - proceed via homolytic fission and free-radical intermediates.
- » **Nucleophilic substitution ($\text{S}_{\text{N}}1$, $\text{S}_{\text{N}}2$)** - a nucleophile replaces a leaving group via heterolytic fission.
- » **Electrophilic addition** - typical of alkenes, where an electrophile attacks the electron-rich double bond.
- » **Elimination** - atoms or groups are removed to create a new double bond.

A representative $\text{S}_{\text{N}}1$ sequence: **Step 1** $\text{R-X} \rightarrow \text{R}^+ + \text{X}^-$ (slow, heterolytic fission); **Step 2** $\text{R}^+ + \text{Nu}^- \rightarrow \text{R-Nu}$ (fast nucleophilic attack).

8.3 Factors affecting bond strength

Factor	Effect on strength	Illustration
Bond length	Shorter bonds are stronger (better overlap)	C-C (154 pm) < C=C (134 pm) < $\text{C}\equiv\text{C}$ (120 pm) in strength order
Bond order	Higher order = stronger, shorter bond	N_2 (order 3) is stronger than O_2 (order 2)
Resonance	Delocalisation stabilises and equalises bonds	Benzene's C-C bonds are all equal
Inductive effect	Electron-withdrawing groups weaken, donating groups strengthen	$-\text{NO}_2$ withdraws; $-\text{CH}_3$ and $-\text{OH}$ donate

Bond order is calculated from molecular-orbital theory as (bonding electrons - antibonding electrons) / 2, tying this section neatly back to Section 6.

§9 FUNCTIONAL GROUPS & DRUG REACTIVITY

A drug molecule can be viewed as a carbon skeleton decorated with **functional groups** - reactive clusters of atoms that give the molecule its personality. Each group carries characteristic reactivity and controls ionisation, solubility in water versus fat (polarity), chemical stability, how the body metabolises the drug, and how it binds its receptor. Because of this, functional groups are decisive for a drug's activity and potency.

9.1 The four functional-group classes

Class	Behaviour	Representative groups
Acidic	Ionise to donate H^+ ; impart water-solubility; form salts with bases; form ionic and H-bonds with targets	Carboxylic acids, phenols, sulphonamides
Basic	Ionise to accept H^+ ; impart water-solubility; form salts with acids; strong H-bonders	Amines, imines, guanidines, hydrazines, N-heterocycles
Neutral	Intermediate polarity; no net charge	Alcohols, amides, esters, ethers, ketones, lactams, lactones, nitro
Lipophilic	Non-polar; favour lipid media; aid membrane crossing	Alkanes, alkenes, alkynes, aromatic hydrocarbons and halides

PHARMACEUTICAL RELEVANCE

The acid-base character of a functional group determines whether a drug is formulated as its free acid/base or as a more soluble **salt** - the reason so many medicines are sold as hydrochlorides, sodium salts, and the like.

9.2 Functional-group quick reference

Class	Group	Prefix / suffix	Example
Alkane	-C-C-	alkyl / -ane	CH ₃ CH ₂ CH ₃
Alkene	-C=C-	alkenyl / -ene	CH ₃ CH=CH ₂
Alkyne	-C≡C-	alkynyl / -yne	CH ₃ C≡CH
Alcohol	-OH	hydroxy / -ol	CH ₃ CH ₂ OH
Aldehyde	-CHO	-al	CH ₃ CH ₂ CHO
Ketone	-C=O	-one	CH ₃ COCH ₃
Amine	-NH ₂	amino / -amine	CH ₃ CH ₂ NH ₂
Amide	-CONR ₂	carbamoyl / -amide	CH ₃ CONH ₂
Carboxylic acid	-COOH	carboxy / -oic acid	PhCOOH
Ester	-COOR	-oate	CH ₃ COOCH ₂ CH ₃
Ether	-O-	-	CH ₃ CH ₂ OCH ₂ CH ₃
Nitrile	-C≡N	cyano / -nitrile	CH ₃ CN
Nitro	-NO ₂	nitro	PhNO ₂
Sulphonic acid	-SO ₃ H	sulpho / -sulphonic acid	PhSO ₃ H

§10 IUPAC NOMENCLATURE OF ORGANIC COMPOUNDS

Chemical nomenclature is the agreed set of rules for naming substances so that a pharmacist in Berlin and a chemist in New York mean exactly the same molecule. Two systems coexist: **trivial (common) names**, which often refer to a source (formic acid from *formica*, the Latin for ant; ascorbic acid from 'anti-scurvy'), and the systematic **IUPAC** names built from a simple logic: **Prefix + Parent + Unsaturation + Suffix**.

10.1 The four steps of IUPAC naming

- » **Step 1 - find the parent chain:** the longest continuous carbon chain that contains the highest-priority functional group.
- » **Step 2 - number the chain:** number from the end that gives the principal functional group (or, failing that, the first double/triple bond or substituent) the lowest possible locant.
- » **Step 3 - identify substituents:** anything branching off the parent chain (methyl, ethyl, chloro, hydroxy...) is located by number; repeats take di-, tri-, tetra-.
- » **Step 4 - assemble the name:** write it as one word, list substituents alphabetically (ignoring di-/tri- when alphabetising), separate numbers from words with hyphens and numbers from numbers with commas.

10.2 Root names by chain length

C atoms	Root	C atoms	Root	C atoms	Root
1	meth-	8	oct-	15	pentadec-

C atoms	Root	C atoms	Root	C atoms	Root
2	eth-	9	non-	16	hexadec-
3	prop-	10	dec-	17	heptadec-
4	but-	11	undec-	18	octadec-
5	pent-	12	dodec-	19	nonadec-
6	hex-	13	tridec-	20	eicos-
7	hept-	14	tetradec-		

10.3 Naming by degree of saturation

Family	Bonding	General formula	Suffix	Example
Alkane	All single bonds	$C_n H_{2n+2}$ ($C_n H_{2n}$ cyclic)	-ane	3-methylhexane
Alkene	Contains C=C	$C_n H_{2n}$	-ene	hex-2-ene; 1,3-butadiene
Alkyne	Contains C≡C	$C_n H_{2n-2}$	-yne	propyne

Additional conventions: the parent chain must include both carbons of a double/triple bond; two or more multiple bonds give -diene or -triene; cyclic compounds take the prefix **cyclo-**; and when double and triple bonds coexist, IUPAC cites the double bond first but numbers from whichever end reaches a multiple bond soonest. A Greek-letter system (α , β , γ carbons counted from the main functional group) is sometimes used for positions.

10.4 Aromatics and real drug names

Benzene derivatives illustrate how substituent prefixes work: methylbenzene (toluene), hydroxybenzene (phenol), chlorobenzene, aminobenzene (aniline). The same rules name everyday medicines:

Drug	IUPAC name
Ibuprofen	2-(4-isobutylphenyl)propanoic acid
Aspirin (acetylsalicylic acid)	2-acetoxybenzoic acid
Paracetamol	N-(4-hydroxyphenyl)acetamide

§11 PHARMACEUTICALLY IMPORTANT ELEMENTS

The regular march of electronic configuration across the periodic table produces predictable group chemistry - and many of those elements and their compounds are pharmaceutically important, whether as active drugs, excipients or analytical reagents. This section surveys the groups with the greatest relevance to pharmacy.

11.1 General reactivity patterns

- » **Group IA (alkali metals)** are extremely reactive, never found free in nature; they form hydrides, and with oxygen give normal oxides (Li), peroxides (Na) or superoxides (K, Rb, Cs). They react vigorously with halogens to give ionic halides.
- » **Group IIA (alkaline earth metals)** are reactive but less so than Group I; being smaller and more highly charged, their ions form complexes more readily. Magnesium in chlorophyll is coordinated by four nitrogen atoms of a porphyrin ring.

- » Reactivity with water rises down a group: Be is inert, Mg reacts with hot water, while Ca, Sr and Ba react with cold water.

11.2 Group-by-group pharmaceutical highlights

Group / element	Pharmaceutical role	Example compounds
Hydrogen	Constituent of water and countless drugs; H_2O_2 is an antiseptic	Purified / sterile water, hydrogen peroxide
Calcium (IIA)	Blood clotting, cardiac function, bone; antacids	Calcium carbonate, chloride, gluconate BP
Aluminium (IIIA)	Slow-acting antacids; suspending / emulsifying agents	Aluminium hydroxide, phosphate, glycinate; bentonite BP
Phosphorus (VA)	Antacids, buffers, dentifrices	Calcium & magnesium phosphates; phosphoric acid buffers
Oxygen (VIA)	Respiratory therapy; oxidising antiseptic	Medical oxygen, hydrogen peroxide
Iodine (VIIA)	Antiseptics; thyroid disorders; expectorants	Iodine solution, KI, NaI, KIO_3
Silver / Gold	Topical antibacterials; anti-rheumatics	Silver nitrate BP; aurothioglucose USP
Manganese (VIIB)	Oxidising and topical antibacterial agent	Potassium permanganate BP (KMnO_4)

NOTE

The **octet rule** and the periodic trends from Sections 3-4 explain these patterns: elements adopt the oxidation states and reactivities that let them reach stable noble-gas configurations, which is why the chemistry runs so predictably down each group.

§12 COORDINATION COMPOUNDS, COMPLEXES & CHELATION

A **coordination compound** forms when a central metal ion accepts lone pairs of electrons from surrounding molecules or ions through **coordinate (dative) bonds** - the bond type introduced in Section 5. These complexes are indispensable in pharmacy, from EDTA anticoagulants to metal-stabilised insulin.

12.1 Core vocabulary

- » **Ligand** - an ion or molecule that donates an electron pair to the central metal. Ligands donate through atoms such as N, O or a halogen.
- » **Denticity** - how many donor atoms a ligand uses: **monodentate** (one, e.g. CN^-), **bidentate** (two, e.g. oxalate), or **polydentate** (many, e.g. EDTA, which binds through four oxygen and two nitrogen atoms).
- » **Coordination sphere** - the non-ionisable central metal plus its bound ligands, written inside square brackets; the coordination number is the number of donor atoms attached.
- » **Werner's coordination number** - the fixed maximum number of groups a given metal can accommodate, determined by steric factors (e.g. 4 for nickel in $[\text{Ni}(\text{NH}_3)_4]^{2+}$).

12.2 Naming coordination compounds

- » Name the ligands first, in alphabetical order, then the central metal ion.
- » Numerical prefixes (di-, tetra-, hexa-) show how many of each ligand are present but do not affect alphabetical order.
- » The metal's oxidation state is given in Roman numerals in parentheses.

Formula	IUPAC name
$K_4[Fe(CN)_6]$	Potassium hexacyanoferrate(II)
$[Ni(CN)_4]^{2-}$	Tetracyanonickelate(II) ion
$[Zn(OH)_4]^{2-}$	Tetrahydroxozincate(II) ion
$[Ni(CO)_4]$	Nickel tetracarbonyl
$[Co(NH_3)_6]Cl_3$	Hexaamminecobalt(III) chloride

12.3 Chelation and its pharmaceutical uses

Chelation is the coordination of a metal by a polydentate ligand, which clamps the metal like a claw (Greek *chele*). If the resulting complex is a stable, water-soluble species, the ligand is acting as a **sequestering agent** - locking a metal away without physically removing it. This underpins several important applications:

- » **EDTA** prevents discolouration in antibiotic, antihistamine, sulphonamide, adrenaline, anaesthetic and barbiturate preparations by tying up trace metals; it also treats urinary calculi, corneal calcium deposits and hypercalcaemia.
- » As an **in-vitro anticoagulant**, citrate, oxalate and EDTA sequester calcium and so block thrombin formation in stored blood.
- » In **lead poisoning**, EDTA salts (and dimercaprol) form an inert, excretable lead chelate.
- » Analytical reagents such as dithizone and dimethylglyoxime form coloured chelates used to detect trace lead, zinc and nickel.
- » Biological chelation is everywhere - the zinc stabilisation of insulin and metal-dependent enzyme reactions are cornerstones of physiology.

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How to use this guide

Work through the sections in order - each builds on the last. Atomic structure (Sections 1-4) explains *why* atoms bond; bonding theory (Sections 5-8) explains *how*; and the applied chapters (Sections 9-12) show that chemistry at work in real organic and inorganic pharmaceutical compounds. Use the tables for rapid revision and the highlighted boxes to flag the points most worth memorising.

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1 Introduction: Drug Quality & the Pharmacist

A pharmacist is, at heart, a custodian of therapeutic substances. Beyond supplying medicines, the pharmacist carries responsibility for guaranteeing that every product they manufacture, compound, dispense, distribute or sell is fit for its purpose. "Fit" here is a multi-dimensional judgement: the medicine must be correct in its **chemical identity, purity, potency, stability, efficacy** and **safety**, and it must be acceptable to the patient who ultimately takes it. Quality is therefore not a single test but a verdict assembled from many quality characteristics.

The way the profession defines and defends quality has shifted dramatically over time. In earlier practice, quality was judged by a handful of checks on the finished article — appearance, colour, odour, identity, hardness, average weight and similar surface properties. Modern pharmaceutical practice treats those checks as necessary but nowhere near sufficient. Today quality is engineered into the entire manufacturing chain through comprehensive quality-assurance systems that begin at raw-material sourcing and end only when the finished product reaches the patient.

In the past — compliance with standard tests	In recent times — stringency in QA
Identification	Dissolution testing
Disintegration time of solid dosage forms	Bioavailability
Assay of active ingredient(s)	Stability
Test for impurities	Toxicological data
Tablet hardness & uniformity	Whole-process quality-assurance systems

1.1 Quality Assurance, Quality Control & GMP

Three related but distinct ideas structure modern pharmaceutical quality. It helps to hold them apart, because examiners frequently ask you to distinguish them.

Quality Assurance (QA) is the wide umbrella. A **quality-assurance system (QAS)** is the creation and operation of the standards, procedures and management systems that together guarantee the quality of a drug. It is how management gives the customer and the wider community a dependable promise of quality, monitoring the whole journey from acquisition of a pharmaceutical raw material to conversion into a finished product placed in the patient's hands. QA is best summarised by the relationship **QA = QC + GMP**.

Quality Control (QC) is a subset of QA focused on testing and verification — confirming that a product meets predefined specifications. It uses analytical techniques (HPLC, spectroscopy and the like) on raw materials, in-process samples and finished products to confirm potency, purity and stability, and to identify and reject defective batches. QC is essentially **reactive**: it detects problems after production has occurred.

- Involves laboratory testing of drugs using analytical techniques.
- Confirms that potency, purity and stability meet the set standards.
- Identifies and rejects defective batches before they reach the market.

Good Manufacturing Practice (GMP) is the set of regulations and guidelines ensuring that drugs are consistently produced and controlled to quality standards. Where QC tests the output, GMP governs the process itself — facility design, equipment, personnel training, documentation and process validation — and it is essentially **proactive**,

preventing errors, contamination and defects as production happens.

Concept	Nature	Focus	Illustration
QA	Umbrella system	Guarantees quality across the whole process	Governs sourcing → finished product
QC	Subset of QA — reactive	Tests & verifies specifications	Testing a paracetamol batch before release
GMP	Process regulation — proactive	Prevents defects during manufacture	Validating a process for reproducibility

PURPOSE OF A QAS — THE "ZERO-DEFECT" GOAL

A quality-assurance system aims at an essentially defect-free product, such that: (1) each tablet contains the labelled amount of active drug within limits; (2) that drug is fully absorbed on administration (the required bioavailability profile); (3) the product is stable with its adjuvants under normal and adverse conditions; (4) it is made from pure ingredients, in an environment free of toxic contaminants, and packed to guarantee efficacy and meet regulatory requirements; and (5) it is genuinely therapeutically effective in the patient.

1.2 Causes of Poor-Quality Products — The "5Ms"

When product quality fails, the root cause almost always traces back to one of five broad input categories, memorably grouped as the **5Ms**. Each contributes roughly a fifth of the risk, and a robust QAS deliberately controls all five.

The 5 Ms	What can go wrong / how to control it
Machines (equipment & facilities)	Must be appropriately designed, constructed, located, clean and suited to the process.
Materials	Starting materials must be of the right quality, from the right source, and stored under suitable conditions.
Men (personnel)	Personnel errors are avoided when staff are well trained, adequately experienced and highly motivated.
Method	Poorly defined operations and the absence of standardised methods introduce error.
Milieu (environment)	A clean, dust-free, spacious, properly designed environment minimises the risk of error.

2 Assays — Definition, Types & Quality Criteria

An **assay** is an investigative procedure for qualitatively assessing and quantitatively measuring the presence, amount, or functional activity of a target entity — the **analyte** — which may be a drug, a biochemical substance, or an organic sample. In other words, an assay answers two questions at once: "is the substance there?" (qualitative) and "how much of it is there?" (quantitative).

2.1 The Five Families of Assay

Assay family	Basis of measurement
Chemical assays	Chemical procedures separating, identifying and quantifying components.
Biological assays (bioassays)	Comparison of biological response in living tissue against a standard.
Microbiological assays	Inhibition of microbial growth relative to a standard antibiotic.
Immunoassays	Selective antigen–antibody binding, read via a label.
Physical assays	Measurement of physical properties of the analyte.

2.2 Characteristics of a Good Assay Method

Whatever the family, a dependable assay method shares a recognisable set of performance characteristics. These are the yardsticks used to validate and compare methods.

Characteristic	Meaning
Sensitivity	How reliably the test detects the substance as a true positive. A more sensitive test has a lower cut-off, detecting minute amounts of analyte.
Specificity	The ability to assess the analyte unequivocally in the presence of components expected to be present, such as impurities and degradants.
Repeatability	Precision under the same operating conditions over a short time interval (intra-assay precision).
Validity	Evidence that the systems, facilities and processes perform their intended jobs adequately and consistently.
Stability	Evidence of how a drug substance's quality varies with time under environmental influences such as temperature, humidity and light.

SENSITIVITY vs SPECIFICITY — DON'T CONFUSE THEM

Sensitivity is about catching what is truly present (few false negatives). Specificity is about not being fooled by look-alikes — measuring the true analyte cleanly even when impurities, excipients or degradants share the sample. A great assay needs both.

3 Errors, Precision & Accuracy in Analysis

Every analytical result carries uncertainty. Understanding where error comes from — and telling precision apart from accuracy — is central to interpreting assay data honestly.

Three broad types of error can occur during an analysis:

Error type	Character & handling
Gross errors	Easily recognised because they involve a major breakdown — samples split, wrong dilutions prepared, or instruments broken/misused. The result is discarded and the analysis restarted from the beginning.
Systematic errors	A consistent bias in one direction, usually from a single flawed step. Results are precise (tightly clustered) but the mean is displaced from the true value.
Random errors	Unpredictable scatter in both directions. Results are widely spread (imprecise); the mean may still land near the truth by chance.

3.1 Worked Example — Four Students, One Batch

A batch of paracetamol tablets is stated to contain 500 mg per tablet; for this example we take 100% of the stated content as the correct answer. Four students each run a spectrophotometric analysis of a tablet extract five times, obtaining these percentages of stated content:

Student	Five repeat results (% of stated content)	Verdict
1	99.5, 99.9, 100.2, 99.4, 100.5	Precise & accurate
2	95.6, 96.1, 95.2, 95.1, 96.1	Precise but inaccurate (systematic error)
3	93.5, 98.3, 92.5, 102.5, 97.6	Imprecise & inaccurate (random error)
4	94.4, 100.2, 104.5, 97.4, 102.1	Imprecise but accurate mean (by chance)

The mean of each set is found with $\text{mean} = (\sum x_i) / N$, where x_i is an individual value and N the number of measurements. Interpreting the four sets teaches the whole lesson:

- **Student 1** — results cluster tightly around 100% with a mean near the true value. The assay is both precise and accurate; every step was carefully controlled.
- **Student 2** — results are tightly clustered but the mean sits below the true value. The assay is precise yet inaccurate: not random scatter but a **systematic error** from probably one faulty step (“precisely inaccurate”).
- **Student 3** — results are widely scattered and the mean is low. The analysis carries **random errors** and was poorly controlled; correcting it needs more work than Student 2’s.
- **Student 4** — results are widely scattered yet the mean happens to fall close to the truth. Only chance separates this from Student 3; an accurate mean here should not be trusted to recur.

PRECISION vs ACCURACY — THE ONE-LINE TEST

Precision = how tightly repeat results agree with each other (tight cluster). Accuracy = how close the mean sits to the true value. You can be precise without being accurate (Student 2), and accidentally accurate without being precise (Student 4). The goal is both (Student 1).

3.2 Common Sources of Imprecision & Inaccuracy

- Incorrect weighing and transfer of analytes and standards.
- Inefficient extraction of the analyte from a matrix (e.g. from tablets).
- Incorrect use of pipettes, burettes or volumetric flasks.
- Measurement on improperly calibrated instrumentation.
- Failure to use an analytical blank.
- Assay conditions chosen that degrade the analyte.
- Failure to allow for, or remove, interference by excipients.

4 Chemical Assays & Physicochemical Techniques

A **chemical assay** analyses a sample material (the analyte) using a defined set of chemical procedures. It is the study of the separation, identification and quantification of the chemical components of natural and artificial materials, and it splits into two complementary modes:

Mode	What it establishes	Typical methods
Qualitative	Which components / physicochemical properties are present	Extraction, distillation, precipitation
Quantitative	How much of a component is present	Measurement by volume or weight of the substance

Modern chemical analysis leans heavily on **physicochemical techniques**, which read a physical signal (usually light) to infer chemical quantity. The main families are photometry (including flame photometry), colorimetry, spectrophotometry, fluorimetry and chromatography.

4.1 Photometry & the Beer–Lambert Basis

When light passes through a coloured solution, certain wavelengths are selectively absorbed, producing an absorption spectrum; the light that is not absorbed is transmitted and gives the solution its colour. Two linked quantities describe this: **transmittance (T)**, the fraction of light passing through, and **absorbance (A)**, related by $A = \log(1/T)$. The **Beer–Lambert law** — that absorbance is proportional to concentration and path length — underlies quantification across photometry, colorimetry and spectrophotometry.

4.2 Flame Photometry

Flame photometry rests on the principle that matter absorbs light at the same wavelength at which it emits. A sample aspirated into a flame emits light at element-characteristic wavelengths, giving a characteristic flame colour whose intensity reports concentration.

Element	Emission wavelength	Flame colour
Calcium (Ca)	622 nm	Orange
Lithium (Li)	670 nm	Red
Sodium (Na)	589 nm	Yellow

- **Advantages:** simple, inexpensive, specific, sensitive even to ppm levels, and useful for both qualitative and quantitative work.
- **Disadvantages:** cannot directly detect inert gases; cannot accurately measure the true concentration of the metal ion in solution; only liquid samples can be used.

4.3 Colorimetry vs Spectrophotometry

In **colorimetry**, colourless compounds are first converted into coloured ones by chemical reactions under defined conditions; the quantity of colour formed is proportional to the quantity of the original compound, and a colorimeter (built on the Beer–Lambert law) reads it. **Spectrophotometry** is the more sophisticated, sensitive cousin: it covers roughly 200–750 nm and, unlike a simple colorimeter, uses a precisely selected wavelength via a manipulated monochromator and a quartz cuvette.

Feature	Colorimetry	Spectrophotometry
Definition	Determines concentration from the colour a solution carries	Measures how much light of a chosen wavelength a substance absorbs
Wavelength	Fixed wavelengths in the visible range	Precisely selected; UV & IR accessible (200–750 nm)
Sample colour	Needs coloured / colour-developed samples	Can use colourless samples
Quantification	Compares colour intensity by primary colour components	Measures absorbance at a specific wavelength
Sophistication	Inexpensive, portable	Sophisticated & more sensitive

• **Colorimetry advantages:** inexpensive, good for quantitative estimation of coloured compounds, easily transportable. **Limits:** colourless compounds, UV/IR regions and specific wavelengths are awkward. **Uses:** protein and glucose estimation, water-quality testing, haemoglobin in blood.

4.4 Fluorimetry

Fluorimetry measures fluorescence emitted by a substance after excitation; intensity of fluorescence is proportional to concentration. It is comparatively inexpensive, specific and sensitive, and is used pharmaceutically to assay adrenaline, cyanocobalamin, riboflavin, morphine and pentobarbitone.

Drug	Excitation wavelength	Emission wavelength
Hydrocortisone	460 nm	520 nm
Nicotinamide	250 nm	430 nm

5 Chromatography — Principles & Techniques

The word chromatography comes from the Greek for "colour" and "write" — literally "to write with colour", after early separations of plant pigments into orange, yellow, blue-green and green bands. It quickly proved general: any mixture, coloured or not, can be separated. Chromatography exploits the **differential affinities** of a sample's components for a **stationary phase** and a **mobile phase**; because components partition differently between the two phases, they travel at different rates and separate.

5.1 The Language of Chromatography

Term	Meaning
Mobile phase (carrier)	The solvent moving through/over the system, carrying the sample.
Stationary phase (adsorbent)	The fixed phase inside the column/paper to which components adsorb.
Eluent	Fluid entering the column.
Eluate	Fluid exiting the column, collected in flasks.
Elution	Washing a compound through the column with a suitable solvent.
Analyte	The mixture whose components must be separated and analysed.
Affinity	How strongly a component interacts with a phase (high affinity = strong interaction).
Adsorption / Desorption	Sticking onto the stationary phase / releasing back into the mobile phase.

THE "LIKE-WITH-LIKE" RULE

Components are more attracted to whichever phase resembles them chemically — polar with polar, non-polar with non-polar. Ions and polar molecules favour a polar mobile phase such as water; when molecules share a functional group (e.g. a mixture of alcohols all bearing –OH), they separate instead by carbon-chain length: longer chains have greater dispersion forces and interact more strongly.

5.2 Paper & Thin-Layer Chromatography (TLC)

In **paper chromatography** the stationary phase is a strip of absorbent paper; a pencil (never ink) line marks the **origin** where the sample and known standards are spotted. The mobile phase sits below the origin and climbs the paper, carrying components upward. Those with strong affinity for the stationary phase move slowly and stay near the origin; those favouring the mobile phase travel further, toward the **solvent front**. Paper chromatography can run **ascending** or **descending**. **TLC** works on the same principle but on a thin adsorbent layer, and is more sensitive (smaller samples) and faster than paper.

Both are analysed with the **retardation factor**, $R_f = (\text{distance solute moves from origin}) / (\text{distance solvent moves from origin})$. R_f is a value between 0 and 1 that helps identify a component qualitatively; paper and TLC are essentially qualitative techniques.

5.3 Column & Gas Chromatography, and HPLC

In **column chromatography** the sample and mobile phase are passed through a column packed with stationary phase; it may run by adsorption or partition, separates a wide variety of mixtures, but is time-consuming, needs plenty of mobile phase and is relatively expensive. **Gas chromatography (GC)** uses a gas (e.g. helium) as mobile

phase and a high-boiling liquid coated on a solid as stationary phase; the time a compound takes to travel through the column to the detector is its **retention time (Rt)**. **HPLC** is a modernised, high-pressure liquid technique with two common modes:

HPLC mode	Stationary phase	Mobile phase
Normal phase	Polar	Less polar
Reverse phase	Non-polar	Polar

HPLC applies to both volatile and non-volatile compounds, offers high resolution and separation capacity, and serves qualitative and quantitative analysis — identifying, quantifying or purifying individual components. Its advantages include speed, reproducibility, high sensitivity and performance, a chemically inert and varied stationary phase, modest mobile-phase demand, early recovery and easy visualisation of separated components. Applications span drug discovery, clinical and forensic analysis, proteomics, drug metabolism, environmental and cosmetic analysis, structural determination and therapeutic drug monitoring.

5.4 Reading Chromatograms: Rt, Peak Area & Calibration

For column techniques (GC, HPLC), the chromatogram shows one peak per component and is read two ways. The **retention time (Rt)** — how long a component spends on the column — identifies it **qualitatively**. The **area under each peak** measures its concentration **quantitatively**, by reading against a **calibration curve** built from standards of known concentration. A quantitative result is only valid when the sample falls within the calibrated range; a peak area beyond the highest standard must be diluted and re-run, or the standards extended.

5.5 Choosing the Technique

Selection hinges on the sample's properties and each instrument's limits. Paper/TLC are cheapest and separate largely by polarity (water-separable, polar samples). GC demands **volatile** samples — non-polar molecules with only dispersion forces vaporise readily and suit GC; it cannot handle semi- or non-volatile compounds and needs frequent calibration. HPLC handles **non-volatile**, soluble molecules and separates by size, charge, molar mass or hydrophilicity, but the mobile phase must be kept scrupulously clean and pumped at very high pressure.

Technique	Mobile phase	Stationary phase	Analysis / read-out	Qual/Quant
Paper	Liquid solvent	Absorbent paper	Rf value	Qualitative
TLC	Liquid solvent	Thin adsorbent layer	Rf value	Qualitative
Gas (GC)	Inert gas	High-b.p. liquid on solid	Rt + peak area	Both
HPLC	Liquid (pumped)	Packed column	Rt + peak area	Both

6 Immunoassay

An **immunoassay** is a bioanalytical method that uses antibodies as reagents to detect or quantify specific analytes through antigen–antibody complex formation. Its power lies in exquisite **specificity** (biological recognition) married to high **sensitivity** — often reaching picogram-per-millilitre levels — which is why it is indispensable in therapeutic drug monitoring, pharmacokinetics, biopharmaceutical QC, anti-drug-antibody detection and clinical diagnostics.

6.1 Principle: Antigen, Antibody, Epitope & Label

Immunoassays exploit an antibody's ability to recognise and bind one specific macromolecule out of a complex mixture. The bound macromolecule is the **antigen**; the precise region on it that the antibody grips is the **epitope**, while the antibody's binding site is its **paratope**. Binding is driven by non-covalent forces — hydrogen bonding, electrostatic, van der Waals and hydrophobic interactions — giving a very high affinity constant ($K_a \approx 10^7$ – 10^{12} L/mol) that lets the assay detect trace analytes in blood, plasma or urine. The four prerequisites are an **antigen**, an **antibody**, the **analyte** and a **label**.

Because the binding event itself is invisible, a **label** converts it into a measurable signal. Common labels include enzymes (HRP, alkaline phosphatase), radioisotopes (^{125}I , ^3H , ^{14}C), fluorescent dyes (FITC, Alexa Fluor, Cy3/Cy5), chemiluminescent compounds (acridinium esters), and colloidal particles (gold, latex). Signal intensity is interpolated against a **dose–response (calibration) curve** — typically sigmoidal, fitted with 4-parameter or 5-parameter logistic models — to yield concentration.

6.2 Competitive vs Non-Competitive Formats

In a **competitive (inhibition) immunoassay**, unlabelled analyte from the sample competes with a fixed amount of **labelled analyte (tracer, Ag*)** for a limited number of antibody sites. The signal is **inversely** related to analyte concentration: more analyte → less tracer bound → lower signal. This format suits small molecules (haptens — drugs, hormones, pesticides, mycotoxins) that carry only one epitope. In a **non-competitive (immunometric)** assay, antibody is present in excess and the signal is **directly** proportional to analyte; these give superior sensitivity, wider range and greater specificity, and suit larger multi-epitope analytes (proteins, peptides).

Feature	Direct ELISA	Indirect ELISA	Sandwich ELISA	Competitive ELISA
Antigen size	Any	Any	Large (multi-epitope)	Small (hapten/drug)
Antibodies used	1 (labelled)	2 (unlabelled + labelled)	2 (capture + detection)	1 (limited amount)
Signal relationship	Direct	Direct	Direct	Inverse
Sensitivity	Moderate	High	Very high	Moderate
Signal amplification	No	Yes	Yes	N/A
Primary use	Antigen detection	Serology / Ab detection	Protein/drug quantification	Hapten detection

Direct ELISA adsorbs antigen to the plate and detects it with an enzyme-conjugated primary antibody — fast and low-background, but with no amplification and costly labelling. **Indirect ELISA** adds an enzyme-labelled secondary antibody against the primary, amplifying signal and offering flexibility (widely used for serodiagnosis).

Sandwich ELISA captures the analyte between two antibodies binding different epitopes, giving exceptional sensitivity and specificity for complex analytes (cytokines, hormones, biopharmaceuticals), at the cost of complexity and a possible high-dose "hook" effect.

6.3 One-Step vs Two-Step Competitive Formats

Parameter	One-step (simultaneous)	Two-step (sequential)
Incubation	Single — sample, tracer & Ab together	Sequential — sample + Ab first, then tracer
Sensitivity	Moderate	Higher (analyte pre-equilibrates alone)
Specificity	Moderate (cross-reactants compete too)	Higher (pre-binding excludes low-affinity cross-reactants)
Assay time	Shorter	Longer
Antibody off-rate	Less critical	Critically important
Best suited for	Screening, rapid POC, lateral flow	TDM, pharmacokinetics, high-sensitivity work

6.4 Homogeneous vs Heterogeneous Methods

A defining operational split is whether bound label must be physically separated from free label before reading. **Heterogeneous** assays require separation (usually a solid phase plus wash steps), because the label signals the same whether bound or free; ELISA and RIA are the archetypes. **Homogeneous** assays need no separation because binding itself modulates the label's signal, making them fast, simple and easily automated — ideal for small analytes such as abused and therapeutic drugs.

Parameter	Homogeneous	Heterogeneous
Separation step	Not required	Required (wash steps)
Phases	Single (solution)	Two (solid + solution)
Assay time	15–60 minutes	2–5 hours
Sensitivity	Moderate (ng–µg/mL)	High–very high (pg–ng/mL)
Matrix interference	Higher susceptibility	Reduced by washing
Automation / throughput	Excellent, very high	Good, high
Analyte size	Best for small molecules	Both small & large
Examples	EMIT, FPIA, CEDIA, TR-FRET, LOCI	ELISA, RIA, CLIA, ECLIA, TRFIA

COMMON HOMOGENEOUS TECHNOLOGIES (KNOW THE ACRONYMS)

EMIT — antibody binding sterically blocks a drug-enzyme conjugate; enzyme activity is directly proportional to drug.
 FPIA — fluorescence polarisation rises when a small tracer is bound to a large antibody; signal is inverse to analyte.
 CEDIA — split β -galactosidase fragments recombine only when the drug frees the enzyme-donor. LOCI & TR-FRET — proximity-driven signals (singlet-oxygen channelling / lanthanide energy transfer) generated only when binding brings partners close.

7 Microbiological Assays & Bioassays

Some substances — particularly antibiotics and certain hormones — are most meaningfully measured by their biological effect rather than a purely chemical read-out. Two related approaches cover this ground: microbiological assays and bioassays.

7.1 Microbiological Assays

A microbiological assay compares the inhibition of microbial growth produced by an antibiotic under test against that produced by known concentrations of a standard preparation of the same antibiotic. Two standard methods are used: the **cylinder-plate (cup-plate)** method, which measures zones of inhibition around wells of antibiotic on a seeded plate, and the **turbidimetric (tube)** method, which measures growth suppression as reduced turbidity in liquid culture.

7.2 Bioassays

A **bioassay** is the comparative assessment of the relative potency of a test compound against a standard, measured on living or biological tissue. It quantifies the amount of active principle in a pharmaceutical or biological material using a suitable biological system. Bioassays come in three types: **direct assays**; **indirect graded assays** (response rises with dose, potency read from a standard response curve — e.g. acetylcholine contracting frog rectus abdominis muscle); and **indirect quantal ("all-or-none") assays** (e.g. digitalis-induced cardiac arrest in guinea pigs, insulin-induced hypoglycaemic convulsions in mice, LD_{50}/ED_{50} comparisons).

Bioassay	Chemical assay
Less precise	More precise
More time-consuming	Less time-consuming
Active constituent & structure not known	Active constituent & structure fully known
More sensitive	Less sensitive

When is a bioassay indicated? When a chemical method is unavailable, too complex, or insensitive to low doses (e.g. histamine); when the active principle is unknown or cannot be isolated/purified (e.g. insulin, posterior-pituitary extract, peptide hormones); to compare potency of a drug from different sources (e.g. cardiac glycosides, catecholamines); and where biological activity cannot be captured chemically (e.g. cis- vs trans-methylphenidate). It also underpins biological standardisation of natural products such as oxytocin, vasopressin, insulin and heparin.

Principles. Compare the test substance with the International Standard preparation of the same agent; determine how much test substance produces the same biological effect as the standard; ensure the activity assayed is the activity of interest; use standard and test with similar pharmacological effects and mode of action; and minimise biological variation (constant conditions; animals of the same species, sex and weight; enough animals to reduce individual-variation error). **Accuracy limits:** within $\pm 20\%$ of true value is good, within $\pm 10\%$ is excellent. **Drawbacks:** biological variation, troublesome and time-consuming, expensive, and less accurate than physicochemical methods.

7.3 Physiological Salt Solutions & Their Uses

Bioassays on isolated tissue require physiological salt solutions that keep the preparation alive and responsive. Composition varies with the tissue (quantities below are per 10 litres, pH 7.3–7.4).

Ingredient	Frog-Ringer	Kreb's	Tyrode	Ringer-Locke	De Jalon	McEwen
NaCl	65 g	69 g	80 g	91.5 g	90 g	76 g
KCl	1.4 g	3.5 g	2.0 g	4.2 g	4.2 g	4.2 g
MgCl ₂ ·6H ₂ O	—	1.1 g	1.0 g	—	—	—
NaH ₂ PO ₄ ·H ₂ O	0.1 g	1.4 g	0.5 g	—	—	1.4 g
NaHCO ₃	2 g	21 g	10 g	1.5 g	5 g	21 g
CaCl ₂	1.2 g	2.8 g	2 g	2.4 g	0.6 g	2.4 g
Glucose	20 g	20 g	10 g	10 g	5 g	20 g
Aerating gas	Air	O ₂ +5% CO ₂	O ₂ or air	Pure O ₂	O ₂ +5% CO ₂	O ₂ +5% CO ₂

Solution	Preparation it suits	Electrolyte	Physiological function
Frog-Ringer	Amphibian tissue	NaCl	Maintains osmolarity
Kreb's	Mammalian/avian skeletal muscle	K ⁺	Nerve conduction, muscle contraction, heart rate & rhythm
Tyrode	Intestine	Ca ²⁺	Contraction
Ringer-Locke	Heart muscle	Mg ²⁺	Neurotransmission; reduces spontaneous activity
De Jalon	Rat uterus	NaHCO ₃ / NaH ₂ PO ₄	Buffer
		Glucose	Nutrient

8 Concentration Units & Determination of Purity

Quantitative assays rest on being able to express concentration precisely and to judge whether a substance is pure. This section gathers the concentration units you must be fluent in and the elemental-analysis method for testing purity.

8.1 Concentration Units

Unit	Definition	Formula
Molarity (M)	Moles of solute per litre of solution	$M = \text{moles of solute} \div \text{litres of solution}$
Molality (m)	Moles of solute per kilogram of solvent	$m = \text{moles of solute} \div \text{kg of solvent}$
Normality (N)	Gram-equivalents of solute per litre of solution	$N = \text{molarity} \times \text{equivalence factor (n)}$
ppm	Milligrams of substance per litre	$1 \text{ ppm} = 1 \text{ mg L}^{-1}$
ppb	Micrograms of substance per litre	$1 \text{ ppb} = 1 \text{ } \mu\text{g L}^{-1}$

MOLARITY vs MOLALITY — THE DISTINCTION EXAMINERS PROBE

Molarity is defined per litre of solution and so shifts slightly with temperature (volume expands); molality is defined per kilogram of solvent and is temperature-independent. Normality scales molarity by how many reactive equivalents each mole provides (e.g. H_2SO_4 has $n = 2$).

8.2 Determining Purity by Elemental Analysis

Elemental analysis tests purity by checking whether the measured elemental composition of a sample matches the theoretical composition of the pure compound. The workflow is:

- Identify the major compound in the sample.
- Determine its chemical formula.
- From the formula, count how many atoms of each element are present.
- Find each element's mass using atomic mass units from the periodic table.
- Add the individual masses to get the mass of the whole molecule.
- Find each element's mass-percent: $(\text{mass of that element} \div \text{mass of whole molecule}) \times 100$.
- Compare against the measured percentages given — a close match indicates a pure sample.

Worked Example — Calcium chloride (CaCl_2)

A CaCl_2 sample is analysed and found to contain 36.12% calcium and 63.91% chlorine. Is it pure?

Atom	Atomic mass (amu)	Number present	Total weight (amu)
Ca	40.08	1	40.08
Cl	35.45	2	70.90
Total			110.98

$$\% \text{Ca} = (40.08 / 110.98) \times 100 = 36.11\% \text{ and } \% \text{Cl} = (70.90 / 110.98) \times 100 = 63.89\%.$$

The theoretical values (36.11% Ca, 63.89% Cl) match the measured values (36.12%, 63.91%) almost exactly, so **the calcium chloride sample is pure.**

PRACTICE PROBLEMS (TRY THESE)

1. A bottle of chloroform (CHCl_3) is analysed and found to contain 80% Cl, 15% C and 5% H. Compute the theoretical percentages from the formula and decide whether the chloroform is pure. 2. A student checks a reagent bottle of silver nitrate (AgNO_3) and measures 63.5% Ag, 8.2% N and 28.3% O. Work out the theoretical composition and judge the purity.

9 Quick-Revision Summary & Glossary

9.1 At-a-Glance Recap

- **Quality:** QA = QC + GMP. QC tests (reactive); GMP controls the process (proactive). Poor quality traces to the 5Ms — Machines, Materials, Men, Method, Milieu.
- **Assays:** five families (chemical, bio, microbiological, immuno, physical). A good method is sensitive, specific, repeatable, valid and stability-indicating.
- **Errors:** gross (restart), systematic (precise-but-biased), random (imprecise scatter). Precision $\neq$ accuracy.
- **Physicochemical:** photometry/colorimetry/spectrophotometry rest on Beer–Lambert ($A = \log 1/T$). Fluorimetry reads emitted light $\propto$ concentration.
- **Chromatography:** differential affinity for mobile vs stationary phase; like-with-like. Paper/TLC $\rightarrow$ R_f (qualitative); GC/HPLC $\rightarrow$ R_t (qual) + peak area on a calibration curve (quant).
- **Immunoassay:** antigen–antibody–epitope + label. Competitive = inverse signal (small haptens); non-competitive/sandwich = direct signal (large analytes). Heterogeneous needs washing; homogeneous does not.
- **Bioassay:** potency vs a standard on living tissue; graded vs quantal; $\pm 10\%$ excellent, $\pm 20\%$ good.
- **Purity:** compare measured elemental % against theoretical % from the formula.

9.2 Glossary of Key Terms

Term	Definition
Analyte	The target substance being detected or measured in an assay.
Assay	A procedure to qualitatively assess and quantitatively measure an analyte.
QA / QC / GMP	Quality assurance (whole-system) = quality control (testing) + good manufacturing practice (process control).
Sensitivity	A test's ability to correctly detect the substance as a true positive.
Specificity	Ability to measure the analyte unequivocally despite other components present.
Precision	How closely repeated measurements agree with one another.
Accuracy	How closely the mean result approaches the true value.
Beer–Lambert law	Absorbance is proportional to concentration and path length; $A = \log(1/T)$.
Mobile / stationary phase	The moving solvent / the fixed phase in chromatography.
Affinity	Strength of a component's interaction with a phase.
R_f (retardation factor)	Distance moved by solute ÷ distance moved by solvent (paper/TLC).
R_t (retention time)	Time a component takes to travel through a column to the detector.
Antigen / Antibody / Epitope	The bound macromolecule / the recognising immunoglobulin / the exact site bound.
Tracer (Ag*)	Labelled antigen used in competitive immunoassays.
Molarity / Molality / Normality	Moles per litre solution / moles per kg solvent / equivalents per litre.

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